

Draw one mode per outer iteration

Indicator available: each mode has probability 0.5; otherwise use exploration.

Exploration: relation-guided search

Sharpness-weighted pair averages $\rightarrow$ variation
Accumulate offspring; deduplicate the candidate pool

Acquisition and quantile filtering

$A_{\text{exp}} = \bar{R} + \lambda_t \bar{U}$
Weight decreases with progress and model error

Greedy batch construction

Acquisition quality + distance from selected decisions

Indicator: relation-guided search

Unweighted pair averages $\rightarrow$ variation
Accumulate offspring; deduplicate the candidate pool

Relation filter $\rightarrow$ indicator reranking

Top relation scores $\rightarrow$ predicted indicator $\hat{I}$
Retain candidates above the indicator quantile

Ordered batch construction

Select by descending predicted indicator

Retained-set and remaining-budget bound: $|S| = \min(n_{\max}, |\mathcal{H}^m|, FE_{\max} - FE)$